

The Gamma Identities

the reflection formula and the Gauss multiplication formula

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The «Dynamic Principle» Program — the hodge-laboratory

Standalone theorem monograph · Монография 3/16

1 Setting

The entire analytic toolkit of the $N = 15/30$ stand consists of two functional identities of the gamma function. Nothing else is needed: all relations among the Γ -products used by the program follow from these two.

Lemma 1 (reflection formula). *For all $z \in \mathbb{C} \setminus \mathbb{Z}$, $\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin \pi z}$.*

Proof. The key is Euler's integral $\int_0^\infty \frac{u^{z-1}}{1+u} du = \frac{\pi}{\sin \pi z}$ ($0 < \operatorname{Re} z < 1$), proved by residues over the keyhole contour: the unique pole $u = -1$ has residue 1; the two sides of the cut give $(e^{i\pi(1-z)} - e^{-i\pi(1-z)}) \int_0^\infty \frac{x^{z-1}}{1+x} dx = 2\pi i$. The substitution $t = \frac{u}{1+u}$ turns the integral into $B(z, 1-z) = \Gamma(z)\Gamma(1-z)$; analytic continuation extends the identity. $\square$

Lemma 2 (Gauss multiplication). *For integers $m \geq 2$ and $mz \notin \mathbb{Z}_{\leq 0}$, $\prod_{k=0}^{m-1} \Gamma(z + \frac{k}{m}) = (2\pi)^{\frac{m-1}{2}} m^{\frac{1}{2}-mz} \Gamma(mz)$.*

Proof. The ratio $Q_m = G_m/H_m$ is 1-periodic and holomorphic: the recurrence $\Gamma(w+1) = w\Gamma(w)$ gives equal logarithmic derivatives. Stirling's formula gives $Q_m \rightarrow 1$ as $\operatorname{Re} z \rightarrow +\infty$ in vertical strips; periodicity spreads the limit; hence $Q_m \equiv 1$. $\square$

Lemma 3 (inventory of relations). *All used relations among $\Omega_{a,b} = B(a/N, b/N)$ follow from the two identities: (1) $a+b = N$: $\Omega_{a,b} = \pi / \sin(\pi a/N)$; (2) $a+b > N$: reduction $\Omega_{a,b} = \frac{N}{a+b-N} \frac{\Gamma(a/N)\Gamma(b/N)}{\Gamma((a+b-N)/N)}$; (3) $a+b < N$: canonical.*

Proof. Point 1 — reflection at $b/N = 1 - a/N$; point 2 — the recurrence $\Gamma(\frac{a+b}{N}) = \frac{a+b-N}{N} \Gamma(\frac{a+b-N}{N})$; point 3 — by definition. $\square$

Protocol data: numbers

The reflection ladder (V6): $\Gamma(k/N)\Gamma(1-k/N) = \pi / \sin(\pi k/N)$ for all k , $N = 15, 30$; maximal relative deviation $7.0 \cdot 10^{-36}$ at $dps = 35$.

Summary of results: summary

Both identities are derived from integral representations and the residue theorem — no external results. The inventory fixes the scope: the full classification of $\mathbb{Q}$ -relations (Rohrlich) is not used — sufficiency is verified by the rank (V7: rank = g).

Verification: reproduction

`python3 laboratory.py` → “Stand N=15/30 → V6”; the plot `fig_gamma.png` (600 dpi) is the numerical portrait of the identity.

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